James Harrell IV

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OBJECTIVE

Computer Engineering student with strong communication and learning skills, demonstrated through technical proficiency in multiple programming languages, game development, and embedded systems/electronics. Thrived in high-stress situations, consistently increasing performance while remaining dedicated, persistent, and collaborative with others. Completed four shipped games and several practical applications; now seeking a Summer 2027 internship in user-interactive experiences and electronic communication systems.

EDUCATION

Georgia Institute of Technology | Atlanta, GA *June 2026 - Present*

Bachelor of Science in Computer Engineering *Expected Graduation, May 2028*

- Activities: Game Development Club, Intramural Basketball

Kennesaw State University | Kennesaw, GA Fall 2026 - May 2026

Bachelor of Science in Computer Engineering, GPA 3.9

- President's List, Fall 2024 and Spring 2026; Dean's List, Spring 2025 and Fall 2025
- Activities: Game Development Club, Intramural Basketball, Social Justice Club

SKILLS

Programming: Python, Java, C#, GDScript

Technical: Linux CLI, Raspberry Pi / embedded systems, SSH & remote debugging, networking (Wi-Fi/LAN configuration), FFMPEG, JSON, Git

EXPERIENCE

Lifeguard | Swim Atlanta & Dynamo, Atlanta/Roswell, GA *May 2022 – June 2026*

- Guard swimmers and maintain safe water chemistry across 8 pool locations, preventing incidents through vigilant supervision and rapid response
- Handle opening/closing duties and customer service for members, working 30-40 hours per week each summer season

Dishwasher / Host | The Marlay House, Decatur, GA *January 2025 – Present*

- Balance host duties (seating, server coordination) and industrial dishwashing to keep kitchen and dining operations running smoothly, 15-20 hours per week year-round

PROJECTS

Video Auto-Captioning Software | *Individual Project*

- Built a Python application that automatically generates and burns captions into video using OpenAI's Whisper library for speech-to-text transcription
- Parsed transcribed audio into structured JSON and used FFMPEG to render fully customizable captions (font, size, color, outline, placement) into video output

Raspberry Pi Embedded Systems Project | *Individual Project*

- Installed and configured Raspberry Pi OS on a Raspberry Pi 5 via card imaging, then configured Wi-Fi hotspot/LAN connectivity and SSH remote access
- Deployed Python and Bash scripts for hardware control and remote debugging, building working knowledge of Linux CLI and embedded system workflows

Game Development — 2 Original Games | *Individual Projects*

- Tanked Up (2D tank battle puzzle game): wrote 2,000+ lines of code delivering 50 unique levels, 8+ AI opponents, online leaderboards, and an infinite mode; attracted 45+ unique players to the web release
- 4 Years Later (Guitar Hero-style rhythm game): wrote 4,000+ lines of code building 8 original songs, a badge/achievement system, and saved high scores; reached 25+ downloads

Procedural Wave Generation in Python and Blender | *Individual Project*

- Applied the Fast Fourier Transform and Blender's ocean physics engine to generate realistic calm and turbulent wave effects for a 2D isometric game